



Project Guidelines
CTADWEBL- ADVANCED WEB PROGRAMMING

Project: *Full-Stack MERN Web Application*

In this project, students will design and develop a secure Full-Stack Web Application using MERN Stack (MongoDB, Express.js, React.js, and Node.js). The application should solve a real-world problem and demonstrate the concepts learned throughout the course, including React components, JSX, props, state management, hooks, routing, forms, CRUD operations, REST APIs, JWT authentication, and web security practices

Students are encouraged to develop systems such as:

- Student Information System
- Inventory Management System
- Online Appointment System
- Event Management System
- Library Management System
- Task Management System
- E-Commerce System
- Learning Management System
- Clinic Management System
- Hotel Reservation System

The project should showcase both frontend and backend development while implementing secure coding practices.

Objectives:

At the end of the project, students should be able to:

1. Develop a full-stack web application using the MERN Stack (MongoDB, Express.js, React.js, and Node.js) to solve a real-world problem.
2. Apply ReactJS concepts such as components, props, state management, hooks, routing, forms, and API integration in building interactive and responsive user interfaces.
3. Implement database operations and RESTful APIs to perform Create, Read, Update, and Delete (CRUD) functionalities.
4. Integrate authentication and security mechanisms, including JWT authentication, input validation, and secure coding practices, to protect web applications.
5. Demonstrate software development skills through system design, testing, debugging, documentation, and project presentation.

Requirements:



Project Guidelines
CTADWEBL- ADVANCED WEB PROGRAMMING

Development Stack

Students must use:

Frontend- ReactJS, React Router, React Hooks, Axios , Bootstrap or Material UI

Backend - Node.js , Express.js

Database - MongoDB

Authentication - JSON Web Token (JWT)

Suggested Minimum System Features:

Students should include:

User Management - User Registration, User Login, Logout Functionality, Password Encryption, JWT Authentication.

Dashboard - Responsive Dashboard, User Profile Page.

CRUD Operations- Add Records, View Records, Update Records, Delete Records.

React Features - React Components, React Router, React Hooks (useState, useEffect), Form Handling, Form Validation.

API Integration - REST API using Express.js, Axios Integration.

Security Features- Protected Routes, Input Validation, Authentication Middleware, Secure API Access.

User Interface: Responsive Design, Navigation Menu, Search Functionality, Pagination.

Scope of Documentation

Chapter 1

- Introduction

1.1 – Project Context

1.2 – Purpose and Description

1.3 – Objectives

- General Objective
- Specific Objectives

1.4 – Scope and Delimitation

1.5 – System Flowchart

Project Submission Requirements

Students must submit the following:

1. Project Documentation (PDF)
2. Source Code Screenshots
3. System Screenshots
4. Complete Project Source Code
5. Project Presentation

Documentation Format

Bond Paper Size: 8.5" x 11" (Letter)

Margin:
Top: 1 inch
Left: 1.5 inch
Right: 1 inch



Project Guidelines
CTADWEBL- ADVANCED WEB PROGRAMMING

Grading Rubric:

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Points
Functionality & Completeness	All required features fully functional	Minor issues	Some incomplete features	Major features missing	/25
MERN Stack Implementation	Proper integration of React, Node, Express, MongoDB	Minor integration issues	Partial implementation	Poor implementation	/20
Authentication & Security	JWT, protected routes, validations fully implemented	Minor security issues	Basic implementation	No security measures	/15
UI/UX Design & Responsiveness	Professional, responsive, user-friendly	Mostly responsive	Basic interface	Poor design	/10
React Concepts & Best Practices	Excellent use of hooks, routing, components	Minor issues	Basic implementation	Poor use of React concepts	/10
Code Quality & Organization	Clean, modular, documented code	Mostly organized	Some disorganized code	Poor structure	/10
Documentation & Presentation	Complete and professional	Minor missing sections	Basic documentation	Incomplete documentation	/5
Innovation & Extra Features	Multiple advanced features implemented	One or two enhancements	Minimal enhancement	No enhancements	/5



Project Guidelines
CTADWEBL- ADVANCED WEB PROGRAMMING

For computation

Score = (Rating ÷ 4) × Maximum Points

Total: 100 Points

Prepared by:	Checked by:	Approved by:
Ms. Alyssa Joy SIT Faculty	Ms. Emilyn A. Donor IT Program Chair	Ms. Marilou N. Jamis Dean
Mr. Christian E. Del Rosario SIT Faculty		